

Trigonometry



The unit circle and radian measure

- 1 Convert 210° to radians.
 - 2 Convert $\frac{5\pi}{4}$ radians to degrees.
 - 3 Find the exact value of $\sin(240^\circ)$.
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Trigonometric identities

- 4 Simplify $\frac{1 - \sin^2 \theta}{\cos \theta}$.
 - 5 Simplify $\tan \theta \cdot \cos \theta$.
 - 6 If $\sin \theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos \theta$ and $\tan \theta$.
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Double angle formulas

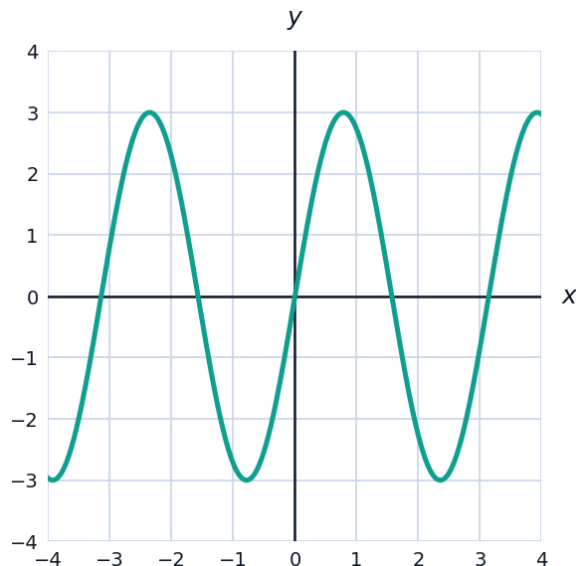
- 7 Given $\cos \theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.
 - 8 Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin \theta$ and $\cos \theta$ only.
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Solving trigonometric equations

- 9 Solve $2\sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
 - 10 Solve $2\cos^2 \theta + \cos \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
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Graphs of trigonometric functions

- 11 The graph shows $y = 3\sin(2x)$. State the amplitude and period.



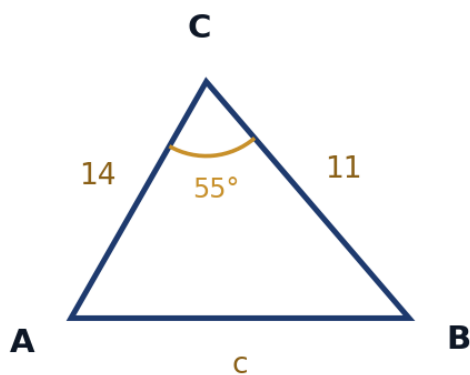
- 12 The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?

The Law of Sines

- 13 In triangle ABC , $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b , to the nearest hundredth.
- 14 In triangle ABC , $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.

The Law of Cosines and the sine area formula

- 15 In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c , to the nearest hundredth, and the area of the triangle, to the nearest tenth.



- 16** State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a , b , c and included angle C .
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Inverse trigonometric functions

- 17** Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.
- 18** Evaluate $\arccos(-1)$ exactly, in radians.
- 19** Find $\theta = \arctan(1)$ in degrees (principal value).
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A well-designed word problem: locating a ship from two lighthouses

- 20** This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.